Aether

Data Analysis Report

Empirical diagnostic evaluation, descriptive statistics, and multivariate exploratory analysis.

DATASET NAME	**2018-2019_Daily_Attendance_20240429.csv**
REPORT DATE	09 October 2026, 18:31 UTC
DATASET VERSION	Version 2
TOTAL RECORDS	277,153 rows
FEATURE DIMENSIONS	6 columns (5 numeric, 1 categorical)
DATA QUALITY SCORE	**100/100**
SCOPE OF ANALYSIS	Full-Spectrum Exploratory & Statistical Audit

Notice: This report is generated dynamically from verified local dataset records. All numerical metrics, distribution profiles, and correlation coefficients are computed strictly from the active dataset version.

Executive Summary

TOTAL RECORDS	TOTAL DIMENSIONS	DATA QUALITY SCORE
277,153	**6**	**100/100**
MISSING DATA RATE	DUPLICATE RECORDS	DETECTED OUTLIERS
0.0%	**0**	**89,545**

This report presents a structured analysis for **2018-2019_Daily_Attendance_20240429.csv** (Version 2). The dataset comprises **277,153 observations** across **6 attributes**. Empirical evaluation yields an overall Data Quality Score of **100/100**, measuring record completeness (100%), uniqueness (100%), and validity (100%).

Key Summary Findings:

• The dataset contains 277,153 verified observations across 6 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'Enrolled' and 'Present' (r = 0.99, strong positive).

• Identified 89,545 statistical outliers using standard IQR thresholds across numeric features.

Dataset Overview

DATASET NAME	**2018-2019_Daily_Attendance_20240429.csv**	ANALYSIS DATE	09 October 2026, 18:31 UTC
ROW COUNT	277,153	COLUMN COUNT	6
FILE FORMAT	CSV	MEMORY FOOTPRINT	25,712.6 KB

Column Overview:

Column	Data Type	Non-null	Missing	Unique	Summary
School DBN	str	277,153	0 (0.0%)	1,583	Top: 01M539 (0%)
Date	int64	277,153	0 (0.0%)	185	[20180904.0 to 20190626.0], Med: 20190129.0
Enrolled	int64	277,153	0 (0.0%)	3,102	[1.0 to 5955.0], Med: 476.0
Absent	int64	277,153	0 (0.0%)	747	[0.0 to 2151.0], Med: 38.0
Present	int64	277,153	0 (0.0%)	3,727	[1.0 to 5847.0], Med: 430.0
Released	int64	277,153	0 (0.0%)	482	[0.0 to 5904.0], Med: 0.0

Data Quality

Quality Dimension	Score	Benchmark	Assessment Detail
Completeness	**100%**	>= 95.0%	Calculates proportion of non-null cells across the observation matrix.
Uniqueness	**100%**	>= 98.0%	Measures absence of identical duplicate row vectors.
Validity	**100%**	>= 90.0%	Evaluates presence of constant zero-variance features.
Consistency	**100%**	>= 85.0%	Checks for type homogeneity and uniform data formatting.

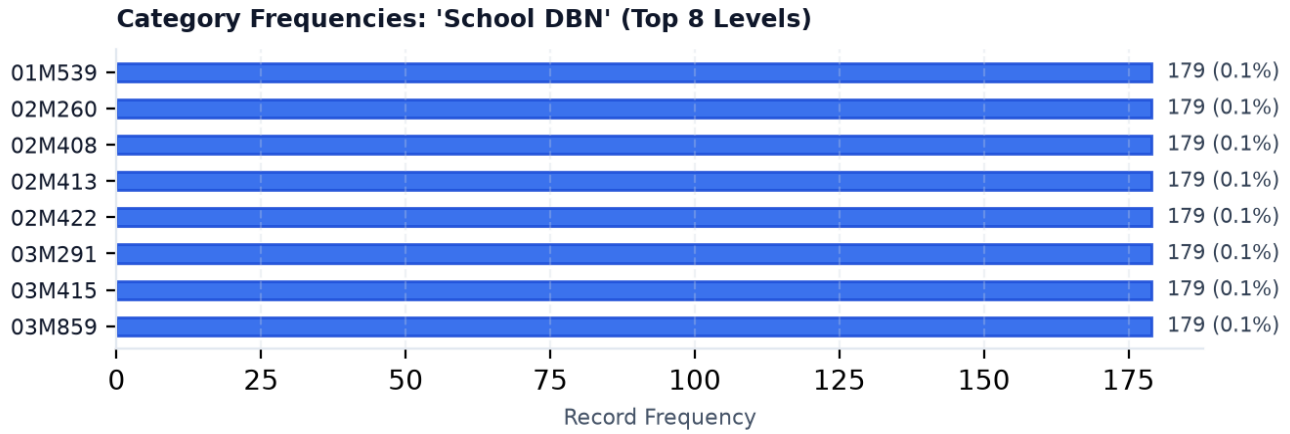
Audit Details: 0 missing cells identified (0.0% of dataset). Duplicate row analysis identified 0 duplicate record(s) (0.0%).

Descriptive Statistics

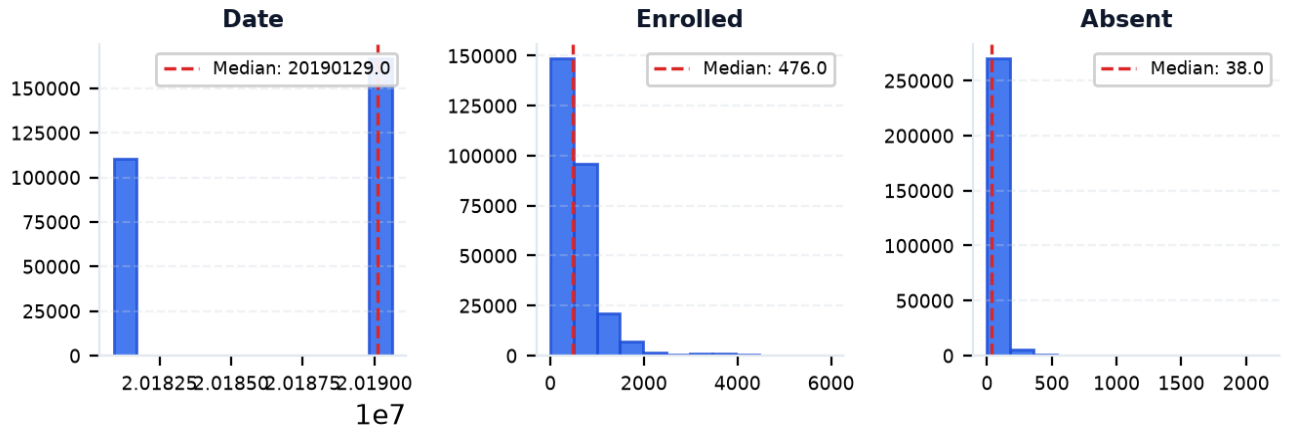
Feature	Count	Mean	Median	Std Dev	Min	Max	Q1	Q3	IQR	Skew
Date	277,153	20186649.1	20190129.0	4555.4	20180904.0	20190626.0	20181113.0	20190410.0	9297.0	-0.41
Enrolled	277,153	597.0	476.0	482.9	1.0	5955.0	329.0	684.0	355.0	3.89
Absent	277,153	50.5	38.0	54.3	0.0	2151.0	23.0	59.0	36.0	6.79
Present	277,153	544.5	430.0	453.0	1.0	5847.0	291.0	640.0	349.0	3.77
Released	277,153	2.0	0.0	35.1	0.0	5904.0	0.0	0.0	0.0	59.74

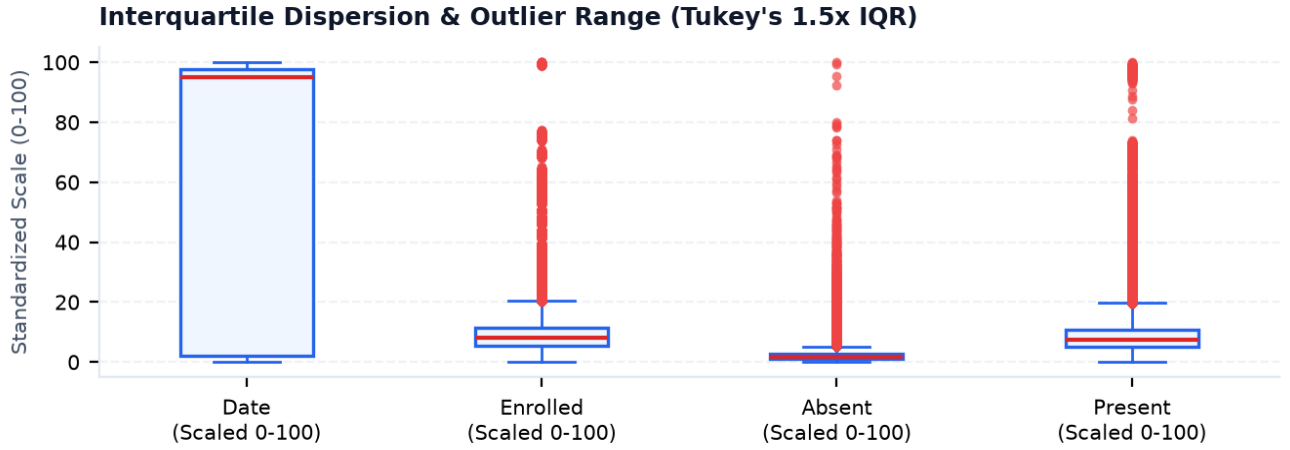
Categorical Analysis

Column	Unique	Most Frequent Value	Frequency	Percentage	Status
School DBN	1,583	01M539	179	0.06%	Balanced

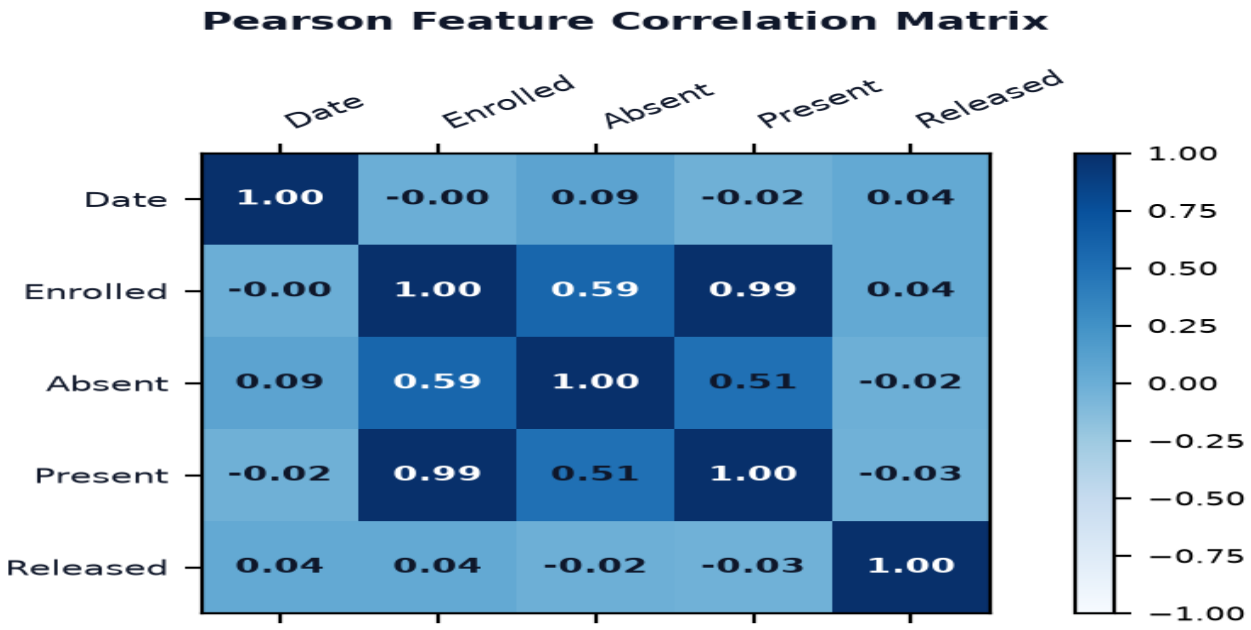


Distribution Analysis



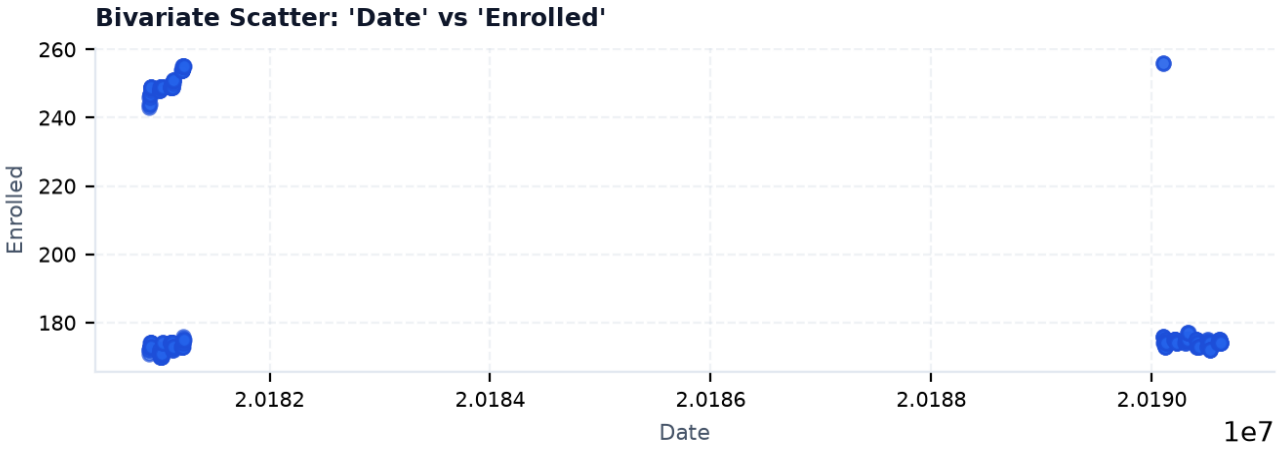


Correlation Analysis



Feature X	Feature Y	Pearson r	Strength	Direction
Enrolled	**Present**	0.99	Strong positive	Positive
Enrolled	**Absent**	0.58	Moderate positive	Positive
Absent	**Present**	0.51	Moderate positive	Positive
Date	**Absent**	0.09	Little or no linear relationship	Positive
Enrolled	**Released**	0.04	Little or no linear relationship	Positive

Relationships



Outlier Analysis

Feature	Outlier Count	Percentage	Lower Bound (1.5x IQR)	Upper Bound (1.5x IQR)
Date	0	0.0%	20167167.50	20204355.50
Enrolled	19,639	7.09%	-203.50	1216.50

Feature	Outlier Count	Percentage	Lower Bound (1.5x IQR)	Upper Bound (1.5x IQR)
Absent	17,555	6.33%	-31.00	113.00
Present	19,051	6.87%	-232.50	1163.50
Released	33,300	12.02%	0.00	0.00

Key Findings

• The dataset contains 277,153 verified observations across 6 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'Enrolled' and 'Present' (r = 0.99, strong positive).

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Recommendations

Category	Priority	Action Recommendation
Analytical	**Medium**	Apply robust scaling or segment upper-quartile outliers in 'Enrolled' to prevent extreme-value distortion.
Analytical	**Medium**	Evaluate 'Enrolled' and 'Present' for potential interaction terms or collinearity redundancy in multivariate regression.
Data Quality	**Low**	Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings.

Appendix

Mathematical Methodology	• **Tukey's IQR Outlier Rule:** Outliers identified strictly outside [Q1 - 1.5*IQR, Q3 + 1.5*IQR]. • **Pearson Correlation:** Linear association computed as r = cov(X, Y) / (std(X) * std(Y)). • **Fisher-Pearson Skewness:** g1 = m3 / (m2^(3/2)) measuring distribution symmetry.
Data Quality Scoring	Overall score computed via weighted evaluation: Completeness (35%) + Uniqueness (25%) + Validity (20%) + Consistency (20%). Penalties scale directly with missing cell frequency and duplicate rows.
Compilation Provenance	Generated by Aether Engine • ReportLab PDF Architecture • Python 3.0.5 / NumPy 2.5.2. Timestamp: 09 October 2026, 18:31 UTC.